

GeNeSIS V - OMNIGENESIS: A Coupled Mathematical Laboratory for Artificial Life

Quantum-Formalism Cognition, Unified Cognitive-Genomic Encoding, Ecology, Chemistry, Evolution, Civilization, and Statistical Inference

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Abstract

GeNeSIS V (OMNIGENESIS) is a from-scratch artificial-life simulation constructed as a coupled discrete-time dynamical system rather than as a collection of independent game mechanics. The implementation combines a 64-dimensional complex cognitive state, agent-specific Hermitian Hamiltonians, Born-style stochastic action selection, an integrated-information-inspired proxy, Hamiltonian-derived genomic encoding, reaction-diffusion morphogenesis, a procedural ecological world, stigmergic pheromone and meme fields, molecular chemistry with Arrhenius kinetics, evolutionary population regulation, social organization, technology formation, cultural transmission, geometric topology, and a broad statistical-analysis layer. The central architectural construction is a shared spectral substrate: the cognitive state dimension is $K = 64$, while the standard four-symbol triplet code has $4^3 = 64$ codons. The system maps the phase structure of Hamiltonian eigenvectors to a 64-symbol nucleotide representation and translates the resulting codons with the standard genetic code. This is an architectural correspondence, not a biological identity claim.

The implementation is deliberately engineered under a low-memory regime. The interactive frontend uses a 56×56 toroidal world, bounded population dynamics, capped histories, immediate pruning of dead agents, procedural terrain generation, lazy panel execution, and a frozen NumPy physics oracle. The quantitative analysis layer includes diversity and inequality indices, spatial statistics, time-series complexity, distribution fitting, survival analysis, information theory, spectral diagnostics, network metrics, inferential tests, signal processing, Granger-style screening, stationarity diagnostics, chaos estimators, statistical-mechanics proxies, and pattern morphology. Built-in tests verify algebraic, numerical, statistical, and geometric invariants using planted reference cases.

The principal contribution is therefore architectural: GeNeSIS V supplies a single executable substrate on which cognitive, genetic, ecological, chemical, evolutionary, cultural, and statistical processes can interact without requiring a separate latent “mind,” “genome,” and “culture” state with manual consistency rules. The manuscript also makes explicit what the system does not establish: the quantum notation is computational formalism on classical hardware; the consciousness quantity is a compact integration proxy rather than a validated theory of phenomenal experience; chemistry is not first-principles molecular dynamics; the physics oracle is synthetic; and long-horizon open-endedness remains an empirical question rather than a proven property.

Keywords: artificial life, agent-based modeling, complex systems, Hilbert-space dynamics, evolutionary computation, reaction-diffusion, stigmergy, statistical inference, computational social science

1 Introduction

Artificial-life systems often fail at the interface between subsystems. Cognition is represented one way, genotype another, environment a third, and social structure a fourth; the resulting simulator then spends substantial complexity synchronizing the representations rather than studying their interaction. GeNeSIS V adopts a different design principle: identify a small set of mathematical objects that can be reused across scales, and make the couplings explicit.

The central construction is a $K = 64$ complex state space. This dimension was inherited from the preceding cognitive architecture, but it becomes structurally important in version V because the standard genetic code also contains exactly 64 codons. The simulator therefore reuses the Hamiltonian eigenbasis as the source of a genome-like symbolic representation. The mapping is intentionally computational. It does not assert that hu-

man biology, silicon computation, or physical quantum mechanics identifies these objects. It asserts a narrower and testable proposition: one representation can be made to support multiple functional roles without introducing a second mutable state variable.

The resulting system couples processes with very different timescales. A cognitive state evolves every tick; stigmergic fields diffuse continuously; role and cellular automaton updates occur at a ten-tick cadence; meme absorption occurs every twenty ticks; viral broadcasting occurs every fifty ticks; reproduction is constrained by an inter-birth cooldown; civilization-level diplomacy is checked at a slower cadence; and the reaction-diffusion field advances multiple substeps per simulation tick. Such scheduling is part of the model because it changes the effective dynamical system.

This paper has four aims. First, it formalizes the principal state variables and transition operators used by the

implementation. Second, it documents the cross-domain couplings that make the system a unified artificial-life laboratory. Third, it describes the quantitative analysis layer as a first-class research instrument rather than a visualization afterthought. Fourth, it separates verified implementation properties from hypotheses that still require controlled experiments.

2 Scientific Position and Scope

The global simulator can be represented schematically by a state vector

$$X_t = \left(W_t, \{A_i(t)\}_{i=1}^{N_t}, C_t, E_t, M_t, T_t, R_t \right), \quad (1)$$

where W_t is the environmental state, A_i is agent state, C_t is civilization state, E_t is evolutionary history, M_t is chemical state, T_t contains time-series histories, and R_t collects derived diagnostics. One tick is a composition of explicit transition maps,

$$X_{t+1} = \mathcal{G} \circ \mathcal{C} \circ \mathcal{E} \circ \mathcal{A} \circ \mathcal{W}(X_t). \quad (2)$$

The operators are implemented in Python modules and connected by an interactive Streamlit frontend.

The system makes several deliberate distinctions. The Hilbert-space machinery is a quantum-formalism model executed on classical CPU hardware. The quantity denoted Φ is an implementation-specific state-integration statistic inspired by Integrated Information Theory, not a complete IIT computation. The chemistry subsystem uses standard constants and reaction equations but does not solve electronic structure or molecular trajectories. Thermodynamic quantities are computed as analogues over simulation variables, and statistical causal diagnostics are screening tools rather than proof of intervention-level causation.

These distinctions are not ancillary caveats. They define the scientific scope of the project and prevent a successful software invariant from being misinterpreted as evidence for a broader physical or philosophical claim.

3 Central Construction: One Spectral Substrate

Let agent i possess a normalized cognitive state

$$\psi_i(t) \in \mathbb{C}^{64}, \quad \psi_i^\dagger \psi_i = 1. \quad (3)$$

Let H_i be a Hermitian Hamiltonian with eigensystem

$$H_i v_{ik} = \lambda_{ik} v_{ik}, \quad k = 0, \dots, 63. \quad (4)$$

The eigenbasis is mapped into a symbolic four-letter alphabet using the phase quadrant of each eigenvector's dominant component. This produces a sequence

$$D_i = d_{i,0} d_{i,1} \dots d_{i,63}, \quad d_{i,k} \in \{A, C, G, T\}. \quad (5)$$

The first sixty-three symbols form twenty-one complete codons, with one remainder symbol:

$$64 = 3 \cdot 21 + 1. \quad (6)$$

The architectural chain is therefore

$$H_i \rightarrow \{v_{ik}\} \rightarrow D_i \rightarrow \text{codons} \rightarrow \text{translated symbols}. \quad (7)$$

A learning update, inherited perturbation, or other change to H_i can alter both future cognitive trajectories and the derived genome-like representation. The key consequence is elimination of a synchronization layer: genotype is a deterministic view of state, rather than an independently mutable object.

4 Mathematical Model

4.1 Cognitive dynamics

The Hamiltonian admits the eigendecomposition

$$H_i = V_i \text{diag}(\lambda_{i,0}, \dots, \lambda_{i,K-1}) V_i^\dagger. \quad (8)$$

The implementation propagates the state through the exact exponential in the eigenbasis,

$$\psi_i(t + \Delta t) = V_i \text{diag}(e^{-i\lambda_{i,0}\Delta t}, \dots, e^{-i\lambda_{i,K-1}\Delta t}) V_i^\dagger \psi_i(t). \quad (9)$$

Because V_i is unitary and each spectral phase has unit modulus, the transformation is norm-preserving in exact arithmetic. Hermiticity is re-imposed after learning updates through the projection

$$H_i \leftarrow \frac{H_i + H_i^\dagger}{2}. \quad (10)$$

These two invariants are explicitly asserted in the implementation tests.

The environment provides a local observation vector formed by four resource channels, sixteen pheromone channels, eight meme channels, and local heat. Thus,

$$\dim(o_t) = 4 + 16 + 8 + 1 = 29. \quad (11)$$

The observation is Fourier-transformed and used to construct candidate action states from the cognitive eigenbasis. For candidate action state $b_a(o_t)$, the action probability has the Born-style form

$$p(a \mid \psi, o_t) \propto |\langle b_a(o_t), \psi \rangle|^2, \quad (12)$$

followed by curiosity-dependent temperature modulation. The procedure is therefore a stochastic policy expressed in quantum-probability coordinates.

4.2 Learning and information-processing cost

The learning operator uses a reward-signed, meta-modulated rank-one Hermitian perturbation. A compact representation is

$$\Delta H = \eta m_t r_t u u^\dagger, \quad (13)$$

where u is an observation/action-derived direction, r_t is the reward term, m_t is the meta-cognitive modulation, and η is the learning scale. The implementation additionally tracks a von Neumann-entropy-like quantity over the

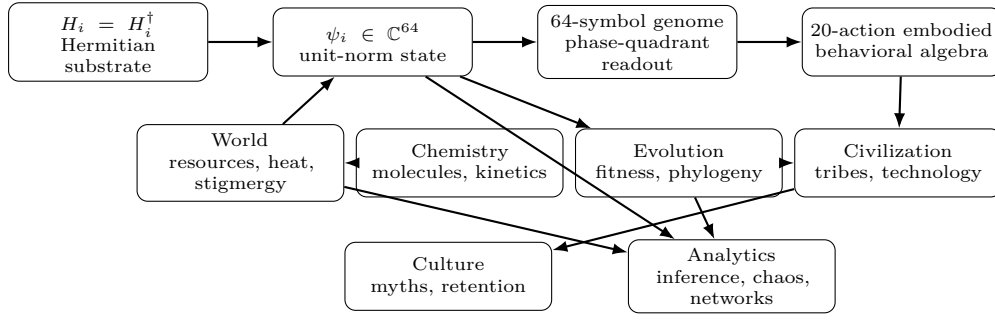


Figure 1: High-level architecture. The central coupling is representational: the Hermitian Hamiltonian and unit-norm state define cognition and genomic readout, while environmental, evolutionary, civilizational, cultural, and analytical subsystems operate on the resulting shared state and histories.

current eigenbasis and charges an information-processing cost proportional to entropy change,

$$c_t = \kappa |\Delta S_t|. \quad (14)$$

This is deliberately a computational analogue of information cost, not a measured thermodynamic heat expenditure.

4.3 Integrated-information-inspired proxy

The 64-dimensional state is partitioned into a 24-dimensional task band and a 40-dimensional complement:

$$\psi = \begin{bmatrix} \psi_{\text{task}} \\ \psi_{\text{comp}} \end{bmatrix}, \quad (24, 40). \quad (15)$$

For a state block x , define

$$I(x) = \log_2 (1 + \text{Var}(|x|)). \quad (16)$$

The implementation computes

$$\Phi = \max(0, I(\psi_{\text{task}}) + I(\psi_{\text{comp}}) - I(\psi)). \quad (17)$$

A sustained increase is flagged when the trailing ten-step mean exceeds 1.2 times a reference mean from the earlier 40-50-step window. The quantity is operationally useful because it is reproducible and longitudinal; it should not be interpreted as a validated consciousness measurement.

4.4 Meta-cognition and symbolic novelty

The behavioral vocabulary contains fixed primitives. Behavioral programs can be represented symbolically and assigned unique Gödel-like integer encodings. In generic form,

$$G(P) = \prod_{j=1}^L q_j^{e_j}, \quad (18)$$

where q_j are fixed primes and e_j encodes primitive/position structure. Novelty is scored by comparing a new program against previously observed programs stored in associative memory. The implementation uses novelty as a reward-shaping signal and supports a five-sigma breakthrough criterion in its novelty scorer.

4.5 Genomic readout and cellular automaton substrate

The Hamiltonian-derived nucleotide string is passed through the standard genetic code. The same derived string seeds an 8×8 Conway Game of Life scratchpad; active cells correspond to two of the four bases and inactive cells to the other two. With $n_t(x, y)$ denoting the local neighbor count, the binary transition is

$$s_{t+1}(x, y) = \begin{cases} 1, & n_t(x, y) = 3, \\ 1, & s_t(x, y) = 1 \wedge n_t(x, y) = 2, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

The grid is toroidal. This is a computational reuse of the genome representation, not a biological model of DNA processing.

4.6 Reaction-diffusion morphogenesis

The morphogenetic field follows the Gray-Scott system:

$$\frac{\partial A}{\partial t} = D_A \nabla^2 A - AB^2 + f(1 - A), \quad (20)$$

$$\frac{\partial B}{\partial t} = D_B \nabla^2 B + AB^2 - (f + k)B. \quad (21)$$

The spatial discretization is a five-point toroidal Laplacian,

$$\nabla^2 u_{i,j} \approx u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}. \quad (22)$$

The frontend advances the morphogen field three substeps per simulation tick. The same spatial operator family is reused for environmental heat diffusion, reducing duplicated numerical machinery.

4.7 Procedural world and ecological fields

The active frontend uses a 56×56 toroidal world. Terrain is generated by a deterministic integer-lattice hash followed by interpolated value noise. A multi-octave field is of the form

$$F(x, y) = \frac{\sum_{o=0}^{O-1} a_o V_o(x, y)}{\sum_{o=0}^{O-1} a_o}, \quad a_o = a_0 p^o, \quad (23)$$

with default persistence $p = 0.5$ and lacunarity 2.0.

Heat obeys an explicit diffusion/forcing step,

$$T_{t+1} = T_t + \alpha \nabla^2 T_t + 0.03 \sin\left(\frac{2\pi t}{200}\right), \quad \alpha = 0.12. \quad (24)$$

The seasonal phase is

$$s(t) = \sin\left(\frac{2\pi t}{200}\right). \quad (25)$$

Resource regeneration couples fertility, temperature, season, weather, and population density. In compact form,

$$\Delta R = 0.04 A_w \frac{1}{1 + \rho_N} (0.5 + 0.5T)(1 + 0.3s)F, \quad (26)$$

with clipping to the configured resource range.

Sixteen pheromone channels and eight meme channels evolve by decay and diffusion,

$$S_{t+1} = \text{clip}(\delta S_t + \gamma \nabla^2 S_t, 0, 10). \quad (27)$$

Pheromones use $(\delta, \gamma) = (0.97, 0.10)$, whereas memes use $(0.995, 0.05)$, creating a deliberately longer cultural memory.

4.8 Frozen PhysicsOracle

The world exposes an eight-dimensional feature vector to a frozen three-layer NumPy MLP. Its map is

$$h_1 = \tanh(xW_1 + b_1), \quad (28)$$

$$h_2 = \tanh(h_1W_2 + b_2), \quad (29)$$

$$y = \tanh(h_2W_3 + b_3), \quad (30)$$

with widths $8 \rightarrow 16 \rightarrow 16 \rightarrow 4$. The weights are generated from a fixed seed and never trained. The oracle therefore provides a deterministic latent law for system-identification experiments without introducing a training loop.

4.9 Embodied agent dynamics

The agent action space contains twenty actions: eight directional moves, eating, attack, communication, reproduction, invention, rest, artifact construction, artifact absorption, meta-invention, composed actions, trade, and punishment. Selected fixed energy costs include

$$c_{\text{move}} = 0.0005, \quad c_{\text{attack}} = 0.20, \quad (31)$$

$$c_{\text{invent}} = 0.15, \quad c_{\text{build}} = 0.05, \quad (32)$$

$$c_{\text{meta}} = 0.06, \quad c_{\text{compose}} = 0.02, \quad (33)$$

$$c_{\text{trade}} = 0.005, \quad c_{\text{punish}} = 0.10. \quad (34)$$

A basal metabolic cost is charged every tick,

$$c_{\text{basal}} = 0.010, \quad (35)$$

and the survival override forces eating below

$$E < 0.4. \quad (36)$$

Reproduction is constrained by a 25-tick cooldown and a population-dependent cost,

$$C_{\text{rep}} = 0.35 \left(1 + 0.5 \left(\frac{N}{128} \right)^2 \right). \quad (37)$$

The model therefore contains explicit resource pressure rather than permitting cost-free existence.

4.10 Roles, artifacts, and cooperation

Roles include Forager, Processor, Warrior, and Queen. At high population, role assignment uses explicit spectral and energetic criteria. Artifact construction instantiates world structures with functional effects: traps damage nearby agents, batteries store energy, and cultivators amplify resource regeneration in a finite radius. Cooperative mega-resources require at least two co-located agents before harvest is permitted, making cooperation a state transition constraint rather than an abstract social label.

4.11 Spectral communication and synchronization

For two agents with Hamiltonian eigenvalue vectors λ_a, λ_b , a spectral-resonance similarity is computed from a cosine-like relation and clipped to a positive floor,

$$\rho(a, b) = \max\left(0.015, \frac{1 + \cos(\lambda_a, \lambda_b)}{2}\right). \quad (38)$$

Communication blends the receiver state toward the sender's phase-weighted state. Population-level phase synchronization follows a Kuramoto-style update every five ticks,

$$\dot{\theta}_i = \omega_i + \frac{\kappa}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad (39)$$

with $\kappa = 0.5$ in the implementation.

4.12 Evolution and population regulation

The evolution engine maintains hard population bounds of 28 and 128. Its purpose is not to simulate a specific natural population but to prevent numerical extinction or runaway growth from dominating the experiment. Fitness is meta-cognitive and behavioral, and the engine additionally records cultural-ratchet statistics and phylogenetic structure.

A useful population-scale order parameter is computed from agent phases,

$$R(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|, \quad 0 \leq R \leq 1. \quad (40)$$

The quantity provides a compact measure of collective phase coherence.

4.13 Civilization and technology

Tribe assignment is driven by sampled spectral resonance. An unaffiliated agent joins the highest-scoring tribe only

when the mean resonance exceeds 0.08; otherwise it founds a new tribe. Tribal social capacity grows with technology,

$$N_{\max}^{\text{tribe}} = 12 + 3|V_{\text{tech}}|. \quad (41)$$

Tribal power is an explicit weighted sum of eight quantities,

$$P_T = 0.30 \sum E_i + 0.20 \sum H_i + 2.5D_T + 0.55|T| \quad (42)$$

$$+ 1.5A_T + 3.0\bar{\sigma}_{\text{meta}} + 5.0\bar{\Phi} + 0.10Q_T, \quad (43)$$

where D_T is discovery count, A_T alliance count, Q_T trade volume, and the barred terms are member means.

Alliances can dissolve when collective meta-Hamiltonian eigenspectra diverge beyond

$$D_{AB} = \|\lambda_A - \lambda_B\|_2 > 48. \quad (44)$$

Cultural assimilation applies

$$H'_i = 0.95H_i + 0.05H_T, \quad (45)$$

followed by Hermitian projection. New technologies form nodes in a directed graph and are linked to nearby inventions by Gödel distance. The global technology multiplier evolves as

$$B_{t+1} = \min(3.0, 1.003B_t). \quad (46)$$

A repository-specific singularity-override criterion requires both

$$n_{\text{novelty}} > 0.55 \quad (47)$$

and

$$|V_{\text{tech}}| \geq 15. \quad (48)$$

This is a computational promotion rule, not a physical claim about technological singularity.

4.14 Cultural transmission

A founding myth is represented by sixteen symbols drawn from the same behavioral primitive vocabulary used for invention and Gödel encoding. Under a Swadesh-style retention model, the nominal per-generation retention is $r_0 = 0.805$, increasing toward 0.955 in the presence of strong local sacred-meme influence. The expected retained count after g generations is

$$\mathbb{E}[L_g] = Lr^g. \quad (49)$$

Fidelity is evaluated with Pearson correlation between founding and current motif representations, thereby separating persistence from exact symbolic identity.

4.15 Chemistry and thermodynamic analogues

The chemistry subsystem distinguishes bulk elemental composition from an explicit molecular-species pool. Reaction rates follow

$$k(T) = A \exp\left(-\frac{E_a}{RT}\right). \quad (50)$$

Reference reactions are checked for atom conservation:

$$\sum_j n_{j,e}^{(\text{reactants})} = \sum_j n_{j,e}^{(\text{products})} \quad \forall e. \quad (51)$$

Statistical-mechanics diagnostics operate over simulation variables. For example, the equipartition estimator with d degrees of freedom is

$$\hat{T} = \frac{2\langle E \rangle}{d}, \quad (52)$$

while the Boltzmann entropy is

$$S = k_B \ln \Omega. \quad (53)$$

The implementation uses natural units in the live dashboard, but these quantities remain computational analogues of their physical counterparts.

5 Quantitative Analysis Layer

The analysis library was expanded into a multi-tier measurement system. The goal is not to maximize the number of metrics, but to cover complementary questions: abundance structure, spatial organization, temporal dependence, inferential comparison, directional predictability, spectral state structure, equilibrium-like distributions, and geometric morphology.

5.1 Diversity, inequality, and spatial structure

Shannon entropy is computed as

$$H = - \sum_i p_i \log p_i, \quad (54)$$

with Simpson concentration $D = \sum_i p_i^2$, inverse Simpson $1/D$, Pielou evenness, Hill numbers, Berger-Parker dominance, and multiple richness estimators. Inequality metrics include Gini, Theil, Atkinson, Hoover, Palma, and Herfindahl-Hirschman indices, together with Lorenz curves.

Spatial structure is characterized by Moran's I , Clark-Evans nearest-neighbor index, quadrat dispersion, Ripley's K and L , and spatial entropy. The intention is to detect clustering, regularity, heterogeneity, and organization that are not obvious from mean population statistics alone.

5.2 Time-series complexity

The implementation includes autocorrelation, Hurst exponent, spectral entropy, permutation entropy, Lyapunov-style proxies, detrended fluctuation analysis, Fano factor, coefficient of variation, Kendall trends, sample and approximate entropy, Lempel-Ziv complexity, Higuchi fractal dimension, recurrence rate, autocorrelation time, Allan variance, and burstiness. These estimators allow a population trajectory to be treated as a dynamical signal rather than merely a chart.

5.3 Statistical inference

For two or more groups, the live statistical panel computes ANOVA, Welch's t test, Mann-Whitney U , two-sample Kolmogorov-Smirnov statistics, permutation tests, Cohen's d , Hedges' g , and Cliff's δ . For example, the standardized mean difference is

$$d = \frac{\bar{x} - \bar{y}}{s_p}, \quad (55)$$

where s_p is the pooled sample standard deviation. Hedges' g applies a small-sample correction.

The system also computes Pearson and Spearman correlations with p-values, chi-square independence tests, bootstrap confidence intervals, and a live energy-distribution comparison between younger and older agents. That age split is median-based: the implementation computes the population median age, partitions agents into $\leq$ median and $>$ median, and applies both KS and Welch tests when both groups contain at least two agents.

Bootstrap uncertainty is estimated using percentile intervals from resampling. Given statistic $\hat{\theta}$ and bootstrap estimates $\hat{\theta}^{*(b)}$, the displayed interval is the empirical percentile range

$$\text{CI}_{1-\alpha} = \left[q_{\alpha/2}(\hat{\theta}^*), q_{1-\alpha/2}(\hat{\theta}^*) \right]. \quad (56)$$

5.4 Signal processing and directional predictability

The analysis layer computes normalized cross-correlation over positive and negative lags. For time series x_t and y_t , a normalized cross-correlation family is formed over shifts ℓ and the lag maximizing absolute correlation is reported. Spectral coherence is estimated using Welch's method.

A Granger-style test compares a restricted autoregression of target y_t against an unrestricted model augmented with lags of source x_t . The comparison can be written as

$$F = \frac{(\text{RSS}_R - \text{RSS}_U)/q}{\text{RSS}_U/(T - k)}, \quad (57)$$

where q is the number of additional source-lag terms. The statistic tests incremental linear predictability, not causal mechanism in the interventionist sense.

Stationarity is screened with an ADF-style regression,

$$\Delta x_t = \alpha + \beta x_{t-1} + \gamma \Delta x_{t-1} + \epsilon_t, \quad (58)$$

with the implementation returning the t statistic for β .

5.5 Chaos and nonlinear dynamics

Two nonlinear diagnostics are central: the Gottwald-Melbourne 0-1 chaos test and a Grassberger-Procaccia-style correlation dimension estimator. The former constructs translation variables from a sequence and estimates a statistic $K \in [0, 1]$; the latter fits the scaling of correlation sums in an embedded trajectory. These are intended as diagnostic evidence, not proof by themselves.

5.6 Spectral and quantum-information diagnostics

Population-wide Hamiltonian analysis includes participation ratio, inverse participation ratio, measurement entropy, spectral gaps, level-spacing ratios, matrix effective rank, eigenvector delocalization, purity, ℓ_1 coherence, state fidelity, trace distance, Bures angle, spectral radius, Frobenius norm, condition number, commutator norm, and matrix effective rank. The live dashboard uses the complex-Hermitian Gaussian Unitary Ensemble reference $\langle r \rangle \approx 0.5996$ and the Poisson reference $\langle r \rangle \approx 0.386$ for level-spacing diagnostics.

These references are used descriptively. A finite simulation of synthetic Hamiltonians cannot be equated with a physical many-body quantum system without additional assumptions and scaling analysis.

5.7 Statistical mechanics and morphology

The thermodynamics panel fits Maxwell-Boltzmann-like speed distributions, estimates temperature by equipartition, computes Boltzmann and Gibbs entropy, a Helmholtz free-energy proxy, entropy-production rates, and a virial-like ratio. A two-dimensional Maxwell-Boltzmann speed density has the form

$$f(v; \sigma) = \frac{v}{\sigma^2} \exp\left(-\frac{v^2}{2\sigma^2}\right), \quad v \geq 0. \quad (59)$$

The dashboard explicitly labels these quantities as analogues because the agent system is not a gas with a physical Hamiltonian.

Morphological diagnostics include circularity, convex-hull solidity, aspect ratio from moments, field orientation, orientation coherence, local-binary-pattern texture entropy, patch statistics, and contagion. The geometry module separately supplies Betti numbers and L-system growth. These descriptors permit the evolving Gray-Scott field to be studied as structured morphology rather than only as an image.

6 Integrated Simulation Cycle

The ordering of updates matters. One high-level tick is:

1. update world heat, seasonal forcing, resources, weather votes, pheromones, and memes;
2. sense local world state and construct an observation vector;
3. propagate the cognitive state and sample an action using Born-style probabilities;
4. apply hunger/fear/integration mode biases;
5. execute the embodied action, producing environmental and social side effects;
6. perform reward-driven Hamiltonian learning and meta-cognitive evolution;
7. perform scheduled synchronization, role updates, Game-of-Life updates, meme/viral transfers;
8. regulate population through the evolution engine;
9. update tribes, civilization memory, diplomacy, schisms, and the technology graph;
10. transmit cultural traditions and evaluate computational breakthrough criteria;
11. advance Gray-Scott morphogenesis through three sub-steps;
12. append bounded history used by the quantitative analysis layer.

This ordering creates path dependence because observations at time t depend on state accumulated from previous

transitions, while the action at time t changes the state supplied to every downstream operator.

7 Software Architecture and Implementation

The current repository is organized around the following modules.

Table 1: Implementation modules and principal mathematical role.

Module	Role
metacognition	Gödel encoding, associative memory, novelty, meta-cognition, phylogeny
consciousness	64D complex state, Hermitian dynamics, Born policy, Φ proxy
biology	Hamiltonian-derived codons, genetic code, Gray-Scott, developmental fields
world	procedural terrain, heat, resources, pheromones, memes, oracle
agents	embodied lifecycle, actions, reproduction, death, communication
evolution	population bounds, meta-fitness, cultural ratchet, archetypes
chemistry	atomic data, molecules, reactions, Arrhenius kinetics
civilization	tribes, power, diplomacy, schism, technology graph
geometry	L-systems, turtle growth, Betti-number topology
narrative	myth formation, retention, fidelity
nobel	six-category computational breakthrough evaluation
analytics	statistical, dynamical, spectral, network, thermodynamic, morphological diagnostics
GeNeSIS V	Streamlit frontend and lazy panel dispatcher

The frontend currently exposes fourteen panels: Observation Deck, Biome Cartography, Consciousness Inspector, Civilization, Narrative, Nobel Committee, Evolution and Phylogeny, Chemistry Lab, Geometry and Topology, Complexity and Dynamics, Ecology and Inequality, Cognitive Spectra, Statistical Inference, and Thermodynamics and Chaos. The panel selector is a sidebar radio rather than native tabs so that expensive rendering remains gated by the currently selected panel.

Memory discipline is architectural. Terrain is procedural rather than globally materialized; simulation history is capped at 400 entries; dead agents are pruned from the live list immediately after their wisdom packets are distributed; and the frontend avoids computing invisible panels. The active world is 56×56 while the world class itself supports larger defaults.

8 Verification and Testing

The project uses layered verification rather than relying only on smoke tests. Module-level tests check mathemati-

cal invariants; cross-module tests exercise interactions; the frontend is additionally driven through Streamlit’s application test framework. The supplied analytics suite contains planted reference cases that are particularly useful because they distinguish an implementation that merely runs from one that recovers known structure.

Table 2: Representative planted verification cases in the analytics layer.

Test	Expected implementation behavior
Near-perfect linear relation	Pearson and Spearman correlations > 0.99 with very small p-values
KS test on same vs. shifted samples	Non-significant p-value for same distribution; strong rejection after location shift
Planted cross-correlation lag	Recovers a known shift of ± 5 ticks
Spectral coherence	Same signal near one; independent noise lower
Granger construction	Planted $X \rightarrow Y$ relation significant; reverse direction non-significant
ADF reference cases	Random walk near unit-root regime; white noise strongly negative
0-1 chaos test	Periodic signal near zero; logistic-map chaos near one
Correlation dimension	Low-dimensional logistic attractor below white noise
Maxwell-Boltzmann recovery	Recovers planted $\sigma = 3$ within the configured tolerance
Equipartition recovery	Recovers planted $T = 4$ within the configured tolerance
Boltzmann entropy	Eight-state construction differs from four-state baseline by $\ln 2$
Morphology bounds	Circularity and solidity never exceed one; concave shapes score lower solidity

The chemistry implementation additionally checks elemental mass balance, and the geometry implementation verifies Betti numbers against known shapes. The cognitive layer checks wavefunction norm preservation and Hamiltonian Hermiticity. The world layer checks finiteness, clipping, deterministic terrain, and cooperative resource gating. Together these tests establish computational correctness for the tested cases; they do not establish that emergent behavior at long horizons is scientifically interesting or representative of natural systems.

9 Scientific Foundations

GeNeSIS V deliberately uses established mathematical formalisms as mechanisms: Turing and Gray-Scott reaction-diffusion for morphogenesis, the French Flag model for positional information, Arrhenius kinetics for temperature-dependent reaction rates, Lotka-Volterra dynamics as an

ecological reference, Kuramoto synchronization for phase coupling, Hopfield-style associative memory, Gödel numbering for symbolic programs, Lindenmayer systems for growth grammars, and standard statistical estimators for population analysis. The implementation is therefore best regarded as a synthesis and computational laboratory rather than a claim that the coupled system constitutes a new fundamental theory.

10 Experimental Protocol for Scientific Use

The current implementation should be treated as an experimental platform. A defensible study using GeNeSIS V should report the random seed, initial population, world dimensions, tick count, population bound configuration, selected analysis metrics, and all relevant calibration constants. At minimum, independent seeds should be treated as experimental replicates rather than pooled as if they were independent agents.

For a treatment/control study, a natural protocol is to disable a single mechanism while keeping all other modules fixed. Candidate ablations include novelty shaping, communication, cultural assimilation, the reproduction cooldown, the metabolic cost, Kuramoto synchronization, or the physics oracle. Let Y_s denote a summary statistic from seed s . A treatment effect can then be represented as

$$\Delta = \frac{1}{S} \sum_{s=1}^S \left(Y_s^{(1)} - Y_s^{(0)} \right), \quad (60)$$

with a seed-level confidence interval or paired resampling procedure. The current dashboard already supplies the ingredients for such summaries, but the repository does not yet contain a preregistered multi-seed benchmark suite; that remains a research task.

For age-group comparisons, the implemented median split produces samples

$$Y_{\leq m} = \{E_i : A_i \leq m\}, \quad Y_{> m} = \{E_i : A_i > m\}, \quad (61)$$

where m is the current population median age. The paired live diagnostics are then

$$D_{KS} = \sup_y |F_1(y) - F_2(y)|, \quad (62)$$

$$T_W = \frac{\bar{E}_1 - \bar{E}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}. \quad (63)$$

The first asks whether the full empirical distributions differ; the second tests a location difference under unequal-variance assumptions. Effect sizes and non-parametric tests should be used alongside p-values when interpreting the panel.

11 Limitations

Several limitations are intrinsic to the current build and should remain visible in any scientific use.

Spectral diversity. Independently initialized agents inherit a highly similar Hamiltonian eigenvalue structure.

The documented result is that pairwise spectral resonance remains high enough that tribe assignment usually collapses into a single tribe. Consequently, diplomacy, alliance, and war mechanisms are implemented but are not reliably activated in current tests. This is a known model calibration finding, not a hidden failure.

Long-horizon dynamics. Current verification covers short to moderate runs, including standalone tests up to roughly hundreds of ticks and frontend exercises to approximately two hundred ticks. The behavior of the coupled system over thousands of ticks has not been established. Correctness of local transitions is therefore not evidence for long-run emergent complexity.

Causal interpretation. Granger-style statistics, correlation, coherence, and lagged association identify statistical dependence or incremental predictability. They do not prove intervention-level causality. Strong causal claims require controlled ablations or interventions on the simulator.

Physical interpretation. The thermodynamic and quantum-statistical panels operate over simulation variables. They should be interpreted as formal diagnostics of the computational system, not as measurements of physical matter, temperature, quantum hardware, or molecular ensembles.

Consciousness claims. The system contains unitary state evolution, Born-style policy selection, an integration statistic, self-reference diagnostics, and information-processing costs. None of these, individually or collectively, establishes phenomenal consciousness.

Open-endedness. The implementation contains novelty search machinery and environment-agent coupling, but the strongest literature definitions of open-ended evolution involve additional mechanisms and very long-horizon evidence. The appropriate claim at present is therefore that GeNeSIS V is a platform for investigating mechanisms related to open-ended behavior.

12 Discussion

The design suggests three broader research questions. First, can representation sharing reduce the engineering burden of artificial-life systems without collapsing distinct phenomena into the same variable? In GeNeSIS V, cognition and genotype share H_i , but their interpretations remain operationally separate. Second, can dense instrumentation change the scientific quality of an agent-based simulator? The analytics layer provides a way to move from visually compelling trajectories to explicit null hypotheses, effect sizes, stationarity checks, lag analysis, and uncertainty intervals. Third, how much emergent structure depends on calibration rather than architecture? The current one-tribe finding illustrates why this question matters: the social mechanism is implemented, but the initial spectral design places the population in a regime where that mechanism is rarely expressed.

The resulting research posture is closer to a mathematical laboratory than to a single benchmark. The simulator

Table 3: Selected formal foundations and their role in the implementation.

Domain	Formalism	Implementation role
Cognition	Hilbert-space dynamics, Born probability	state evolution and action sampling
Information integration	IIT-inspired proxy	longitudinal state-integration diagnostic
Morphogenesis	Turing / Gray-Scott	reaction-diffusion field
Development	French Flag model	positional-information abstraction
Chemistry	Arrhenius kinetics	molecular reaction-rate law
Ecology	Lotka-Volterra and related statistics	population interaction reference
Synchronization	Kuramoto model	collective phase coupling
Associative memory	Hopfield-style dynamics	civilization memory
Symbolic computation	Gödel numbering	program identity and novelty
Growth grammar	Lindenmayer systems	procedural structures
Cultural transmission	Swadesh-style retention	motif persistence model
Statistical analysis	standard inferential and dynamical estimators	population and time-series diagnostics

exposes multiple levels of description over a shared state: micro-level complex amplitudes, agent actions, population statistics, spatial fields, molecular species, symbolic culture, network topology, and civilization-level events. The scientific opportunity is to use these levels together while retaining the ability to isolate each mechanism by controlled ablation.

13 Conclusion

GeNeSIS V presents a unified artificial-life architecture centered on one deliberate computational correspondence: a 64-dimensional cognitive Hilbert space doubles as the source of a genome-like symbolic representation. Around this core, the system integrates reaction-diffusion morphogenesis, procedural ecology, stigmergic communication, molecular chemistry, evolutionary regulation, synchronization, social organization, cultural transmission, technology graphs, topology, and a substantial statistical-analysis layer.

The primary achievement is not a claim of artificial sentence, biological realism, or proven open-ended evolution. It is the construction of an executable environment in which these hypotheses can be probed using explicit mathematics and quantitative diagnostics. The most important next step is therefore experimental rather than architectural: independent multi-seed studies, mechanism ablations, long-horizon trajectories, and pre-specified summary statistics should determine which components produce reproducible emergent phenomena.

Reproducibility Statement

The implementation is written in Python 3.11+ and uses NumPy, SciPy, pandas, Plotly, NetworkX, scikit-learn, and Streamlit. No GPU runtime is required. The frontend is designed for a low-memory environment and uses lazy panel execution. A fixed seed can reproduce the procedural world, initial population placement, and deterministic oracle initialization; stochastic analysis routines expose explicit seeds where applicable. The repository’s module-

level self-tests are executable with each module’s main entry point, while the integrated frontend is exercised through Streamlit application tests.

A Frontend Research Panels

The current interface is organized into fourteen research views. Their purpose is summarized in Table 4.

Table 4: Current interactive research panels.

Panel	Primary measurements / visualization
Observation Deck	population, tribes, technology, mean Φ , bloom epoch
Biome Cartography	meme stigmergy, Gray-Scott patterning, elemental projection
Consciousness Inspector	Φ history, emotion, codons, Game of Life
Civilization	tribes, power, technology graph, diplomacy
Narrative	myth transmission and fidelity
Nobel Committee	six-category computational breakthrough ledger
Evolution and Phylogeny	archetypes, cultural ratchet, clades
Chemistry Lab	elemental and molecular diagnostics
Geometry and Topology	L-systems, Betti numbers, morphology
Complexity and Dynamics	time-series complexity and population-technology coupling
Ecology and Inequality	diversity, richness, inequality, Lorenz structure
Cognitive Spectra	Hamiltonian spectra, participation, spacing, coherence
Statistical Inference	ANOVA, effect sizes, bootstrap CIs, correlations, age distributions
Thermodynamics and Chaos	MB fits, entropy, chaos, ADF, lag/coherence/Granger tests

B Analytics Inventory

The analysis module currently exposes the following classes of estimators:

- diversity and inequality: Shannon, Simpson, Hill, Pielou, Berger-Parker, richness estimators, Gini, Lorenz, Theil, Atkinson, Palma, Hoover, HHI;
- spatial statistics: Moran’s I , nearest-neighbor, quadrat dispersion, Ripley’s K/L , spatial entropy, Geary’s C , patch and contagion statistics;

- temporal complexity: autocorrelation, Hurst, spectral entropy, permutation entropy, Lyapunov proxy, DFA, Fano factor, entropy measures, Lempel-Ziv, Higuchi dimension, recurrence and burstiness;
- distributions and survival: power-law MLE/KS, Zipf slope, Gompertz-Makeham, Kaplan-Meier, distribution shape, Benford deviation;
- information and spectra: mutual information, KL/JS divergence, transfer-entropy proxy, von Neumann and measurement entropy, purity, coherence, fidelity, trace distance, Bures angle, spectral gap and spacing metrics;
- networks and ecology: degree summaries, small-world sigma, logistic growth, trend, centrality, global efficiency, degree-power-law diagnostics;
- statistical inference: Cohen's d , Hedges' g , Cliff's δ , bootstrap intervals, permutation tests, ANOVA, chi-square, Pearson/Spearman, KS, Mann-Whitney, Welch;
- signal and chaos: cross-correlation, best lag, spectral coherence, Granger-style F , ADF-style statistic, 0-1 chaos, correlation dimension;
- statistical mechanics and morphology: Maxwell-Boltzmann fit, equipartition temperature, Boltzmann/Gibbs entropy, Helmholtz proxy, entropy production, virial ratio, circularity, solidity, aspect ratio, orientation, texture entropy.

C References

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